

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458882.8253 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.08 +/- 0.034
Transit depth (Rp/Rs)^2	0.63 +/- 0.54 [%]
Orbital Inclination (inc)	81.01 +/- 2.1
Airmass coefficient 1 (a1)	1.1863 +/- 0.0058
Airmass coefficient 2 (a2)	-0.1231 +/- 0.0049
Scatter in the residuals of the lightcurve fit is	0.49 %
Best Comparison Star	None
Optimal Aperture	4.56
Optimal Annulus	8.19
Transit Duration (day)	0.039 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.08 $\pm$ 0.034 vs 0.1594 (deviation 2.34 σ)
Duration fit vs published	0.039 d vs 0.0512 d (deviation 1.02 σ)
Mid-time O-C	4.3 min
Depth SNR	2.4
Residual scatter	0.49 %

Light curve

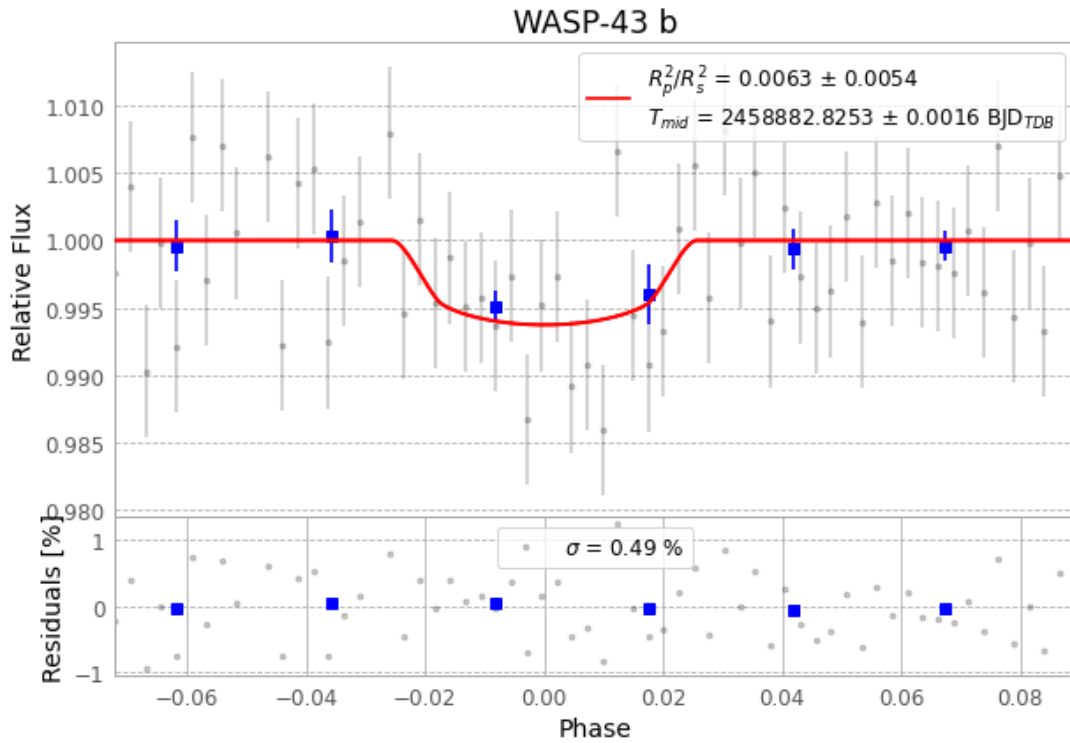


Figure 1 — FinalLightCurve_WASP-43 b_2020-02-02.png (SHA-256 c37eaaa20253dd9d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [255, 334], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c53e8dd6aa2a0bc7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

64 FITS frames (42 MB), WASP-43200203061512.FITS → WASP-43200203092412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-200203.log (b082a805bb61...), FinalLightCurve_WASP-43 b_2020-02-02.png (c37eaaa20253...), FinalLightCurve_WASP-43 b_2020-02-02.csv (80ee13fb48fc...)

Compute resources

Wall time 126.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:34Z.